Berlin citizen watering
Atlas of preliminary maps

Master's Seminar examination | Joseph Carpenter Sheils | 17 October 2026

Twelve existing analytical map plates. Measures, periods, source tables and builders are catalogued in examination/atlas/catalogue.csv. Original image pixels and classifications are preserved. Zoom for labels or open the corresponding PNG.

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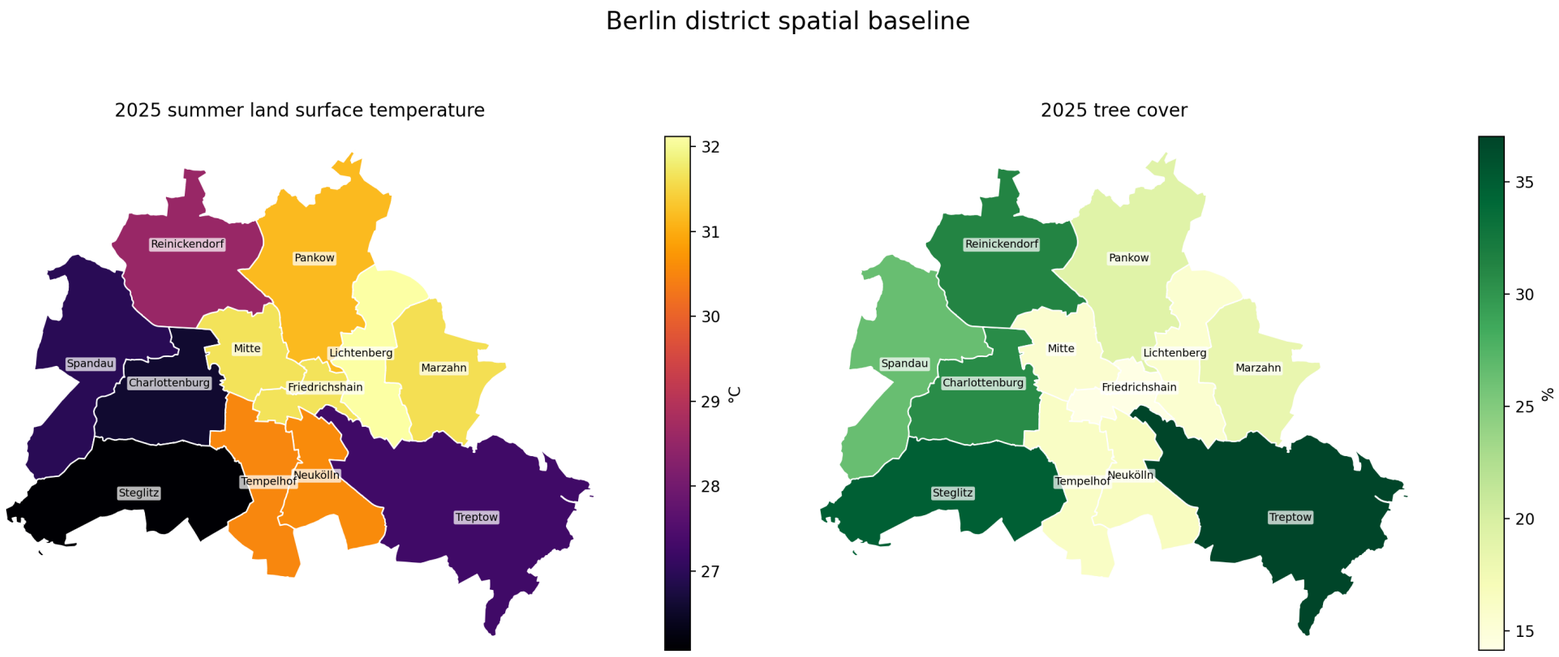
Reading the atlas

Recorded zeros mean no linked platform log during an observed period. They do not establish absence of irrigation. Current-inventory denominators are not historical risk sets. Local Moran classes use unadjusted significance; care-gap candidates require validation. The district baseline mixes source years. No plate is a completed daily response, vegetation cooling-loss, or municipal allocation result.

Attribution: CityLAB / Technologiestiftung Berlin (Gieß den Kiez); Berlin Senate tree, MSS and environmental sources; Amt für Statistik Berlin-Brandenburg, LOR 2021 (CC BY 3.0 DE). Detailed sources, licences and unresolved legacy provenance: data/README.md and data/spatial/README.md. Analysis geometry: EPSG:25833.

01 District spatial baseline

2025 LST and tree cover; May 2020-Dec 2025 watering | Legacy district LST and canopy context; watering reach (%) and litres per inventory tree



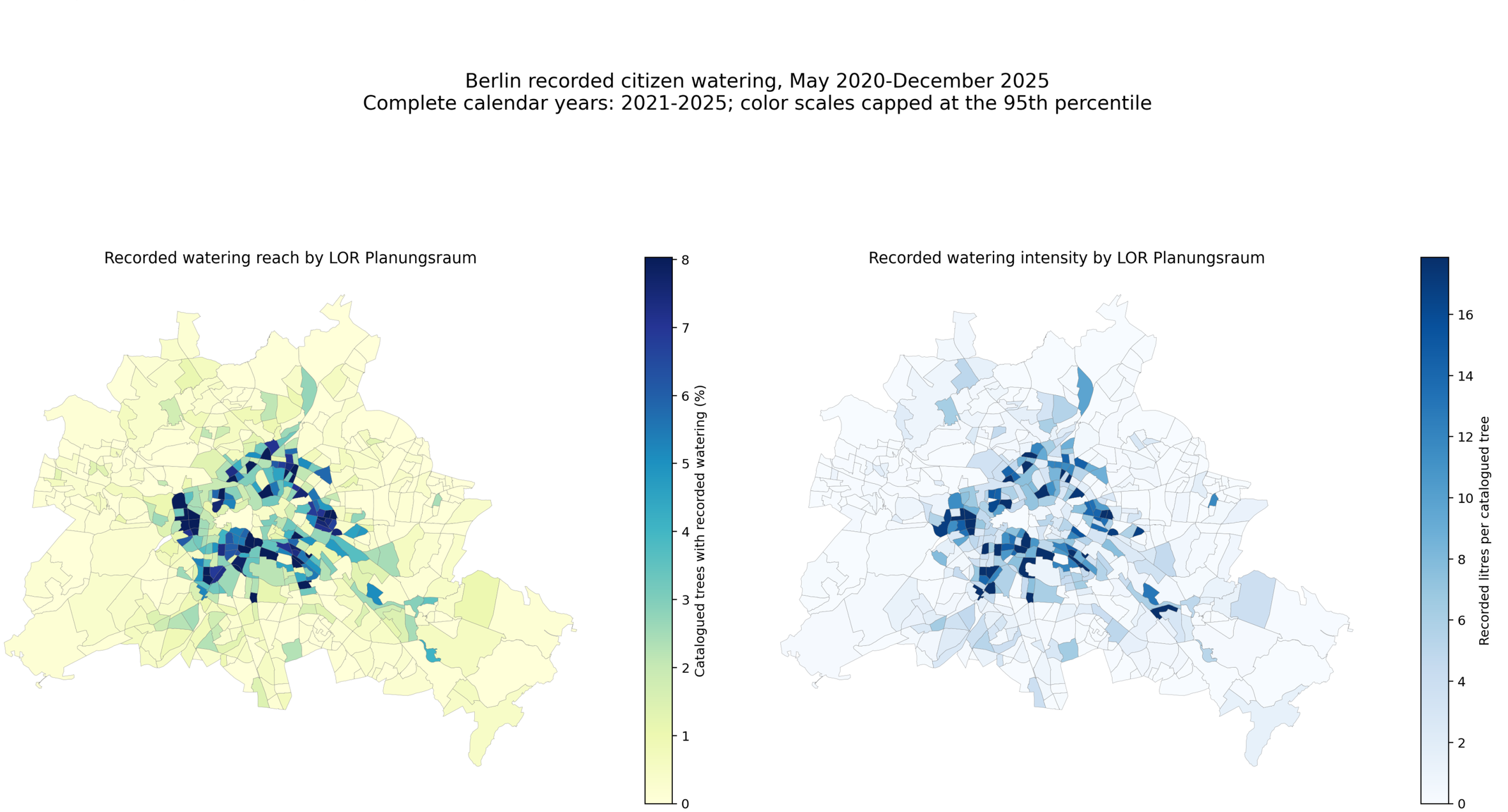
LOR metrics are aggregated by district and joined to the 2025 summer rows of data/Berlin_heat_population_death_Birth_data.csv. Reference periods differ; legacy heat provenance is incomplete. Recorded platform activity only; current-inventory denominators; no inference about unlogged or municipal watering.

Data: output/spatial/tables/lor_watering_metrics_2020_2025.csv

Builder: analysis/build_spatial_notebook.py | Original: output/spatial/figures/01_district_spatial_baseline.png

02 Neighbourhood watering reach and intensity

May 2020-Dec 2025 | Distinct-tree reach (%) and recorded litres per inventory tree



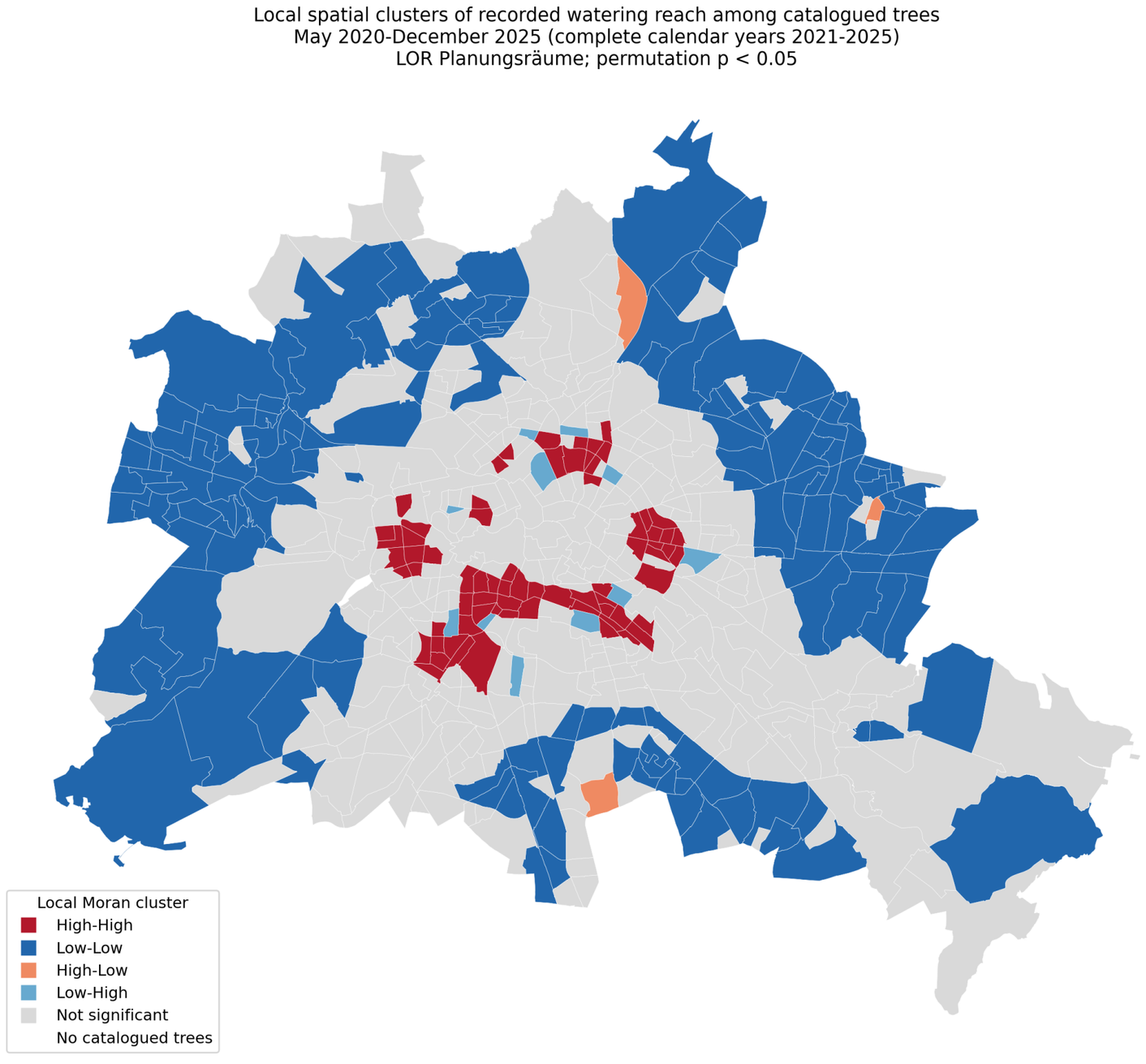
Colour scales are capped at the 95th percentile; inspect the table for extreme values. Recorded platform activity only; current-inventory denominators; no inference about unlogged or municipal watering.

Data: output/spatial/tables/lor_watering_metrics_2020_2025.csv

Builder: analysis/build_spatial_notebook.py | Original: output/spatial/figures/02_lor_watering_reach_intensity.png

03 Exploratory local spatial clusters

May 2020-Dec 2025 | Local Moran classes for recorded watering reach



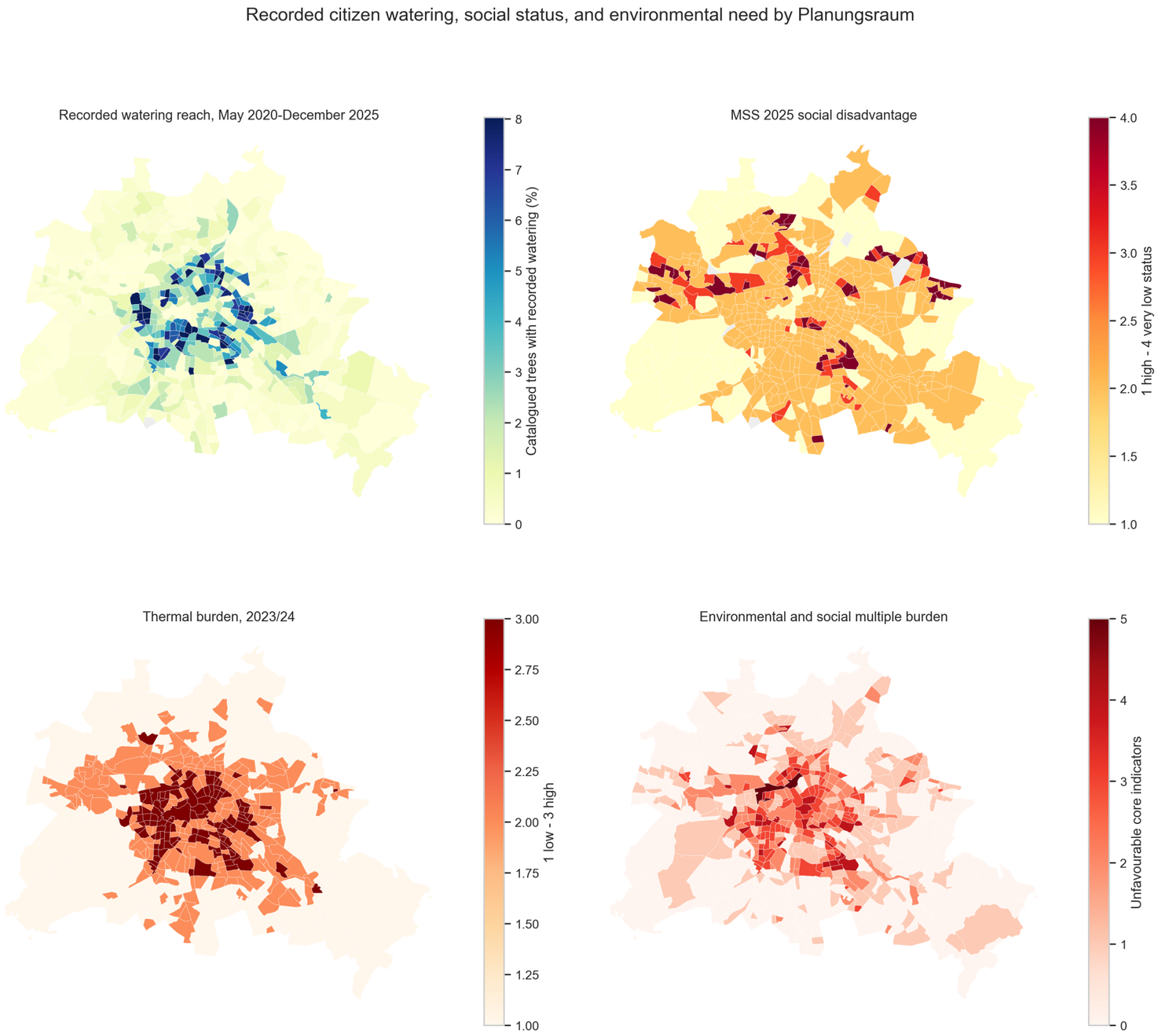
Row-standardised queen contiguity; 999 permutations; unadjusted local p < 0.05, seed 42. Exploratory screening with no multiplicity correction. Recorded platform activity only; current-inventory denominators; no inference about unlogged or municipal watering.

Data: output/spatial/tables/local_moran_watering_clusters.csv

Builder: analysis/build_spatial_notebook.py | Original: output/spatial/figures/03_lor_local_moran_clusters.png

04 Recorded care and social/environmental context

Watering 2020-2025; MSS end-2024; environmental context 2023/24 | Recorded reach, MSS status, thermal burden and environmental/social burden count



542 polygons; 535 valid MSS observations with inventoried trees. Static contextual layers are not historical area-level vulnerability. Recorded platform activity only; current-inventory denominators; no inference about unlogged or municipal watering.

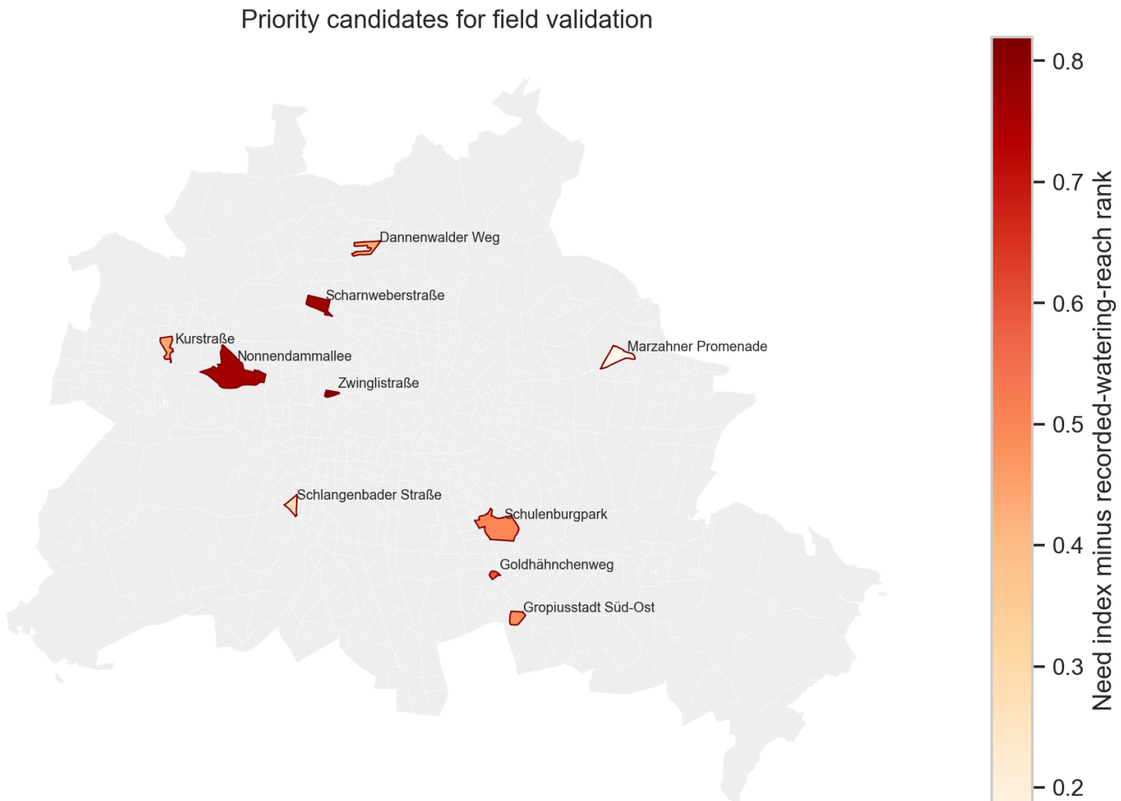
Data: output/equity/tables/lor_watering_equity_metrics.csv

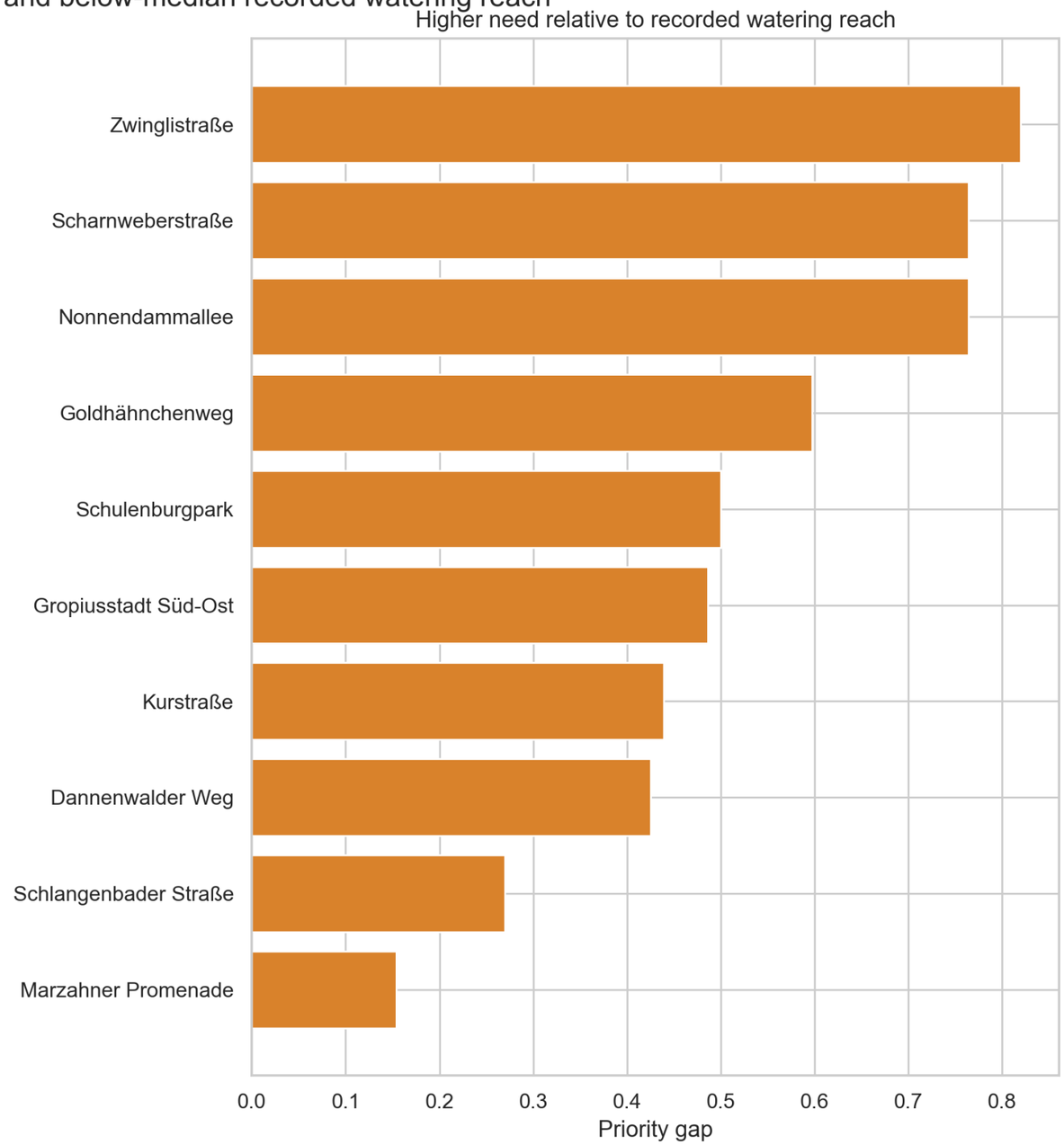
Builder: analysis/build_equity_notebook.py | Original: output/equity/figures/01_lor_watering_social_environment.png

05 Areas for further care-gap validation

Watering 2020-2025; contextual indicators 2023/24 and end-2024 | Low/very-low-status areas with thermal or green need and below-median recorded reach

Low-status LORs with thermal or green need and below-median recorded watering reach





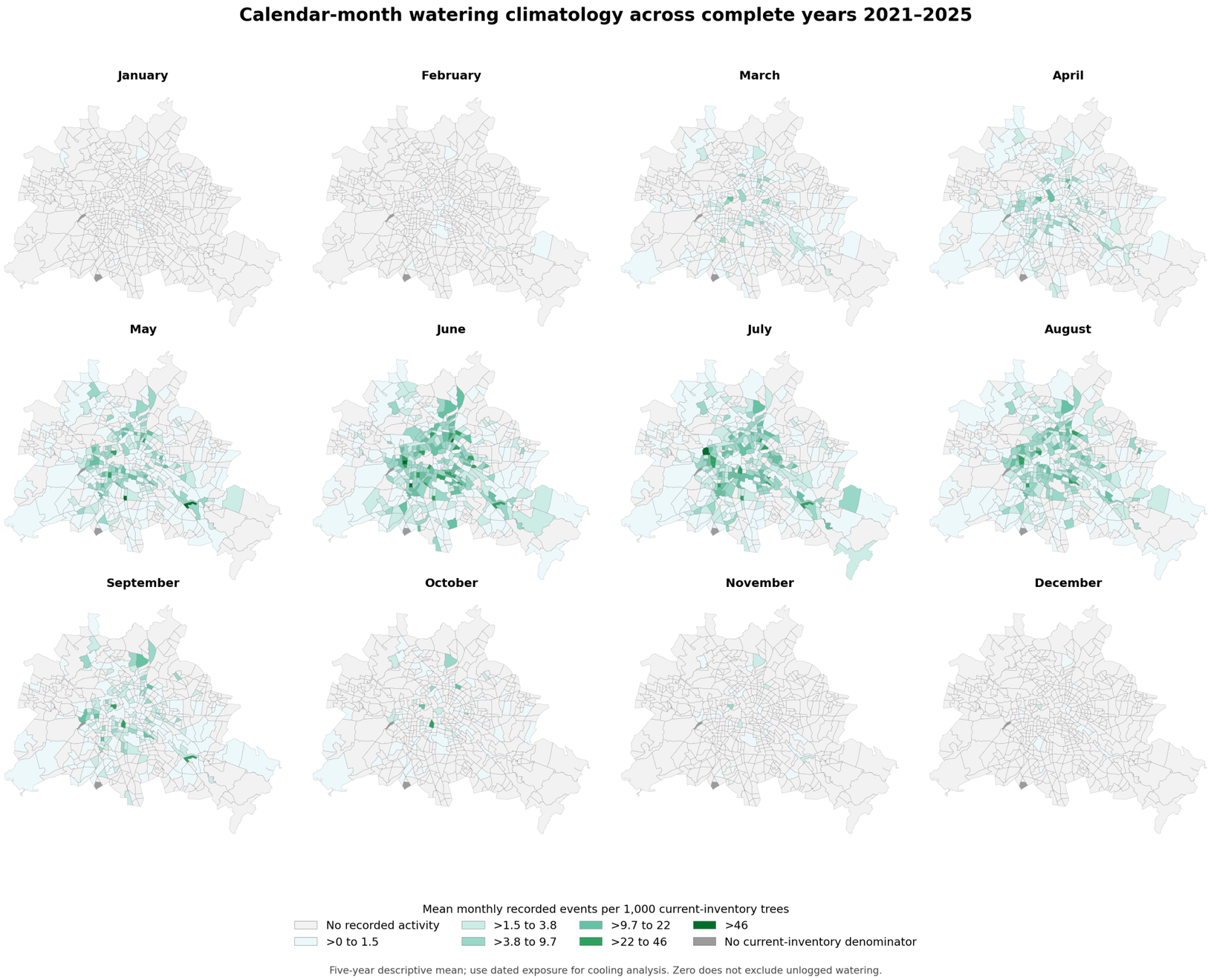
Screening candidates only, not validated irrigation priorities or the proposed allocation model. Recorded platform activity only; current-inventory denominators; no inference about unlogged or municipal watering.

Data: output/equity/tables/priority_gap_planungsraeume.csv

Builder: analysis/build_equity_notebook.py | Original: output/equity/figures/03_priority_gap_candidates.png

06 Calendar-month climatology

Complete years 2021-2025 pooled by calendar month | Mean monthly events per 1,000 current inventory trees



Five-year calendar-month means include observed zeros. Shared fixed classes; no pooled-month thermal inference. Recorded platform activity only; current-inventory denominators; no inference about unlogged or municipal watering.

Data: data/processed/temporal/lor_calendar_month_recorded_watering_2021_2025.csv

Builder: pipelines/04_build_watering_temporal_atlas.py | Original: output/03_spatial_equity_vulnerability/maps/01_calendar_month_recorded_watering_atlas.png

07 Active-season comparison across years

April-September separately in 2021, 2022, 2023, 2024 and 2025 | Monthly events per 1,000 current inventory trees



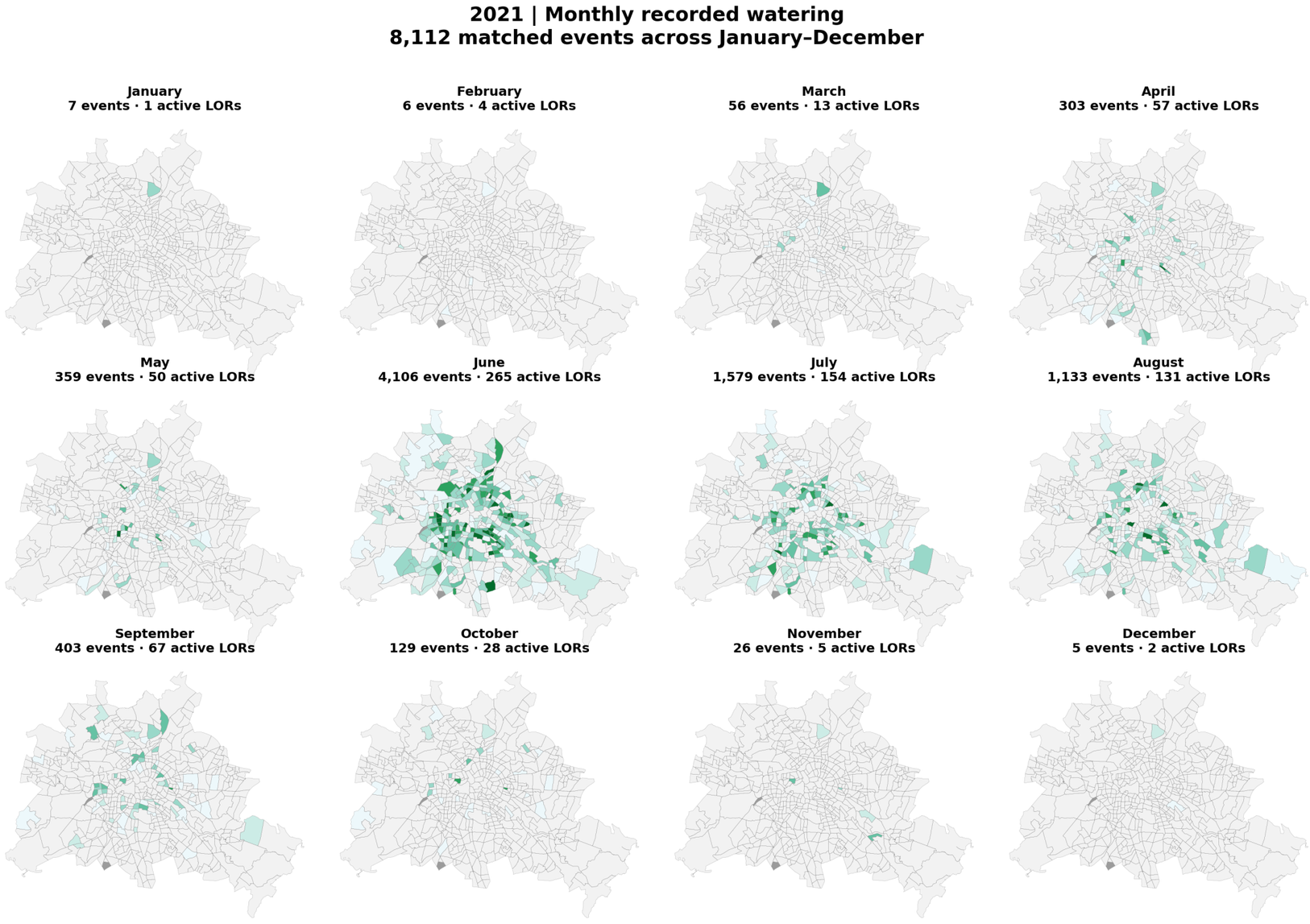
Each panel uses its actual year/month and fixed classes. Recorded platform activity only; current-inventory denominators; no inference about unlogged or municipal watering.

Data: data/processed/foundation/lor_month_recorded_watering_2020_2025.csv

Builder: pipelines/04_build_watering_temporal_atlas.py | Original: output/03_spatial_equity_vulnerability/maps/02_active_season_recorded_watering_atlas.png

08 2021 monthly recorded watering

January-December 2021 | Monthly events per 1,000 current inventory trees



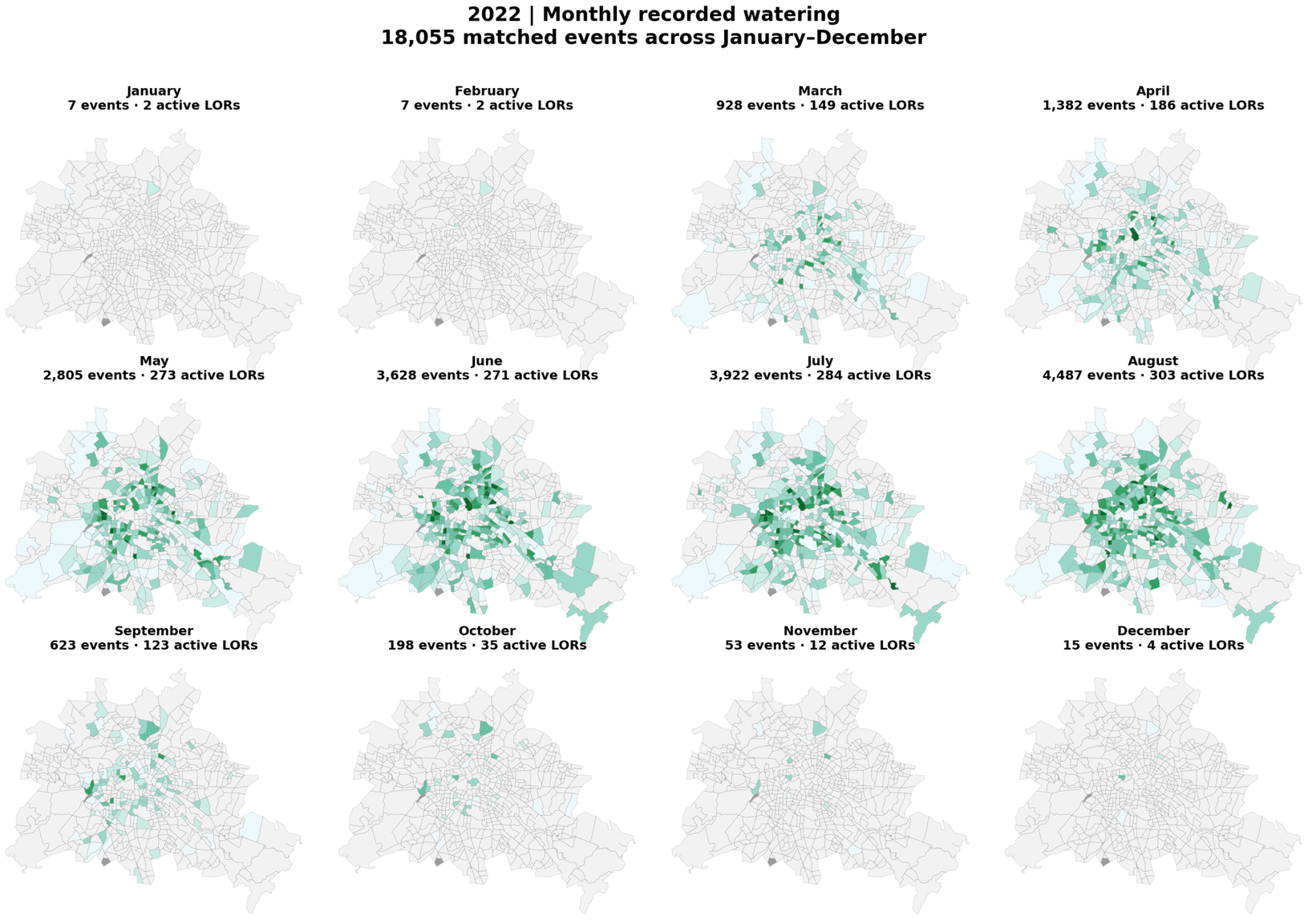
Twelve actual months; shared fixed classes across all years. Recorded platform activity only; current-inventory denominators; no inference about unlogged or municipal watering.

Data: data/processed/foundation/lor_month_recorded_watering_2020_2025.csv

Builder: pipelines/04_build_watering_temporal_atlas.py | Original: output/03_spatial_equity_vulnerability/maps/yearly/2021_monthly_recorded_watering_atlas.png

09 2022 monthly recorded watering

January-December 2022 | Monthly events per 1,000 current inventory trees



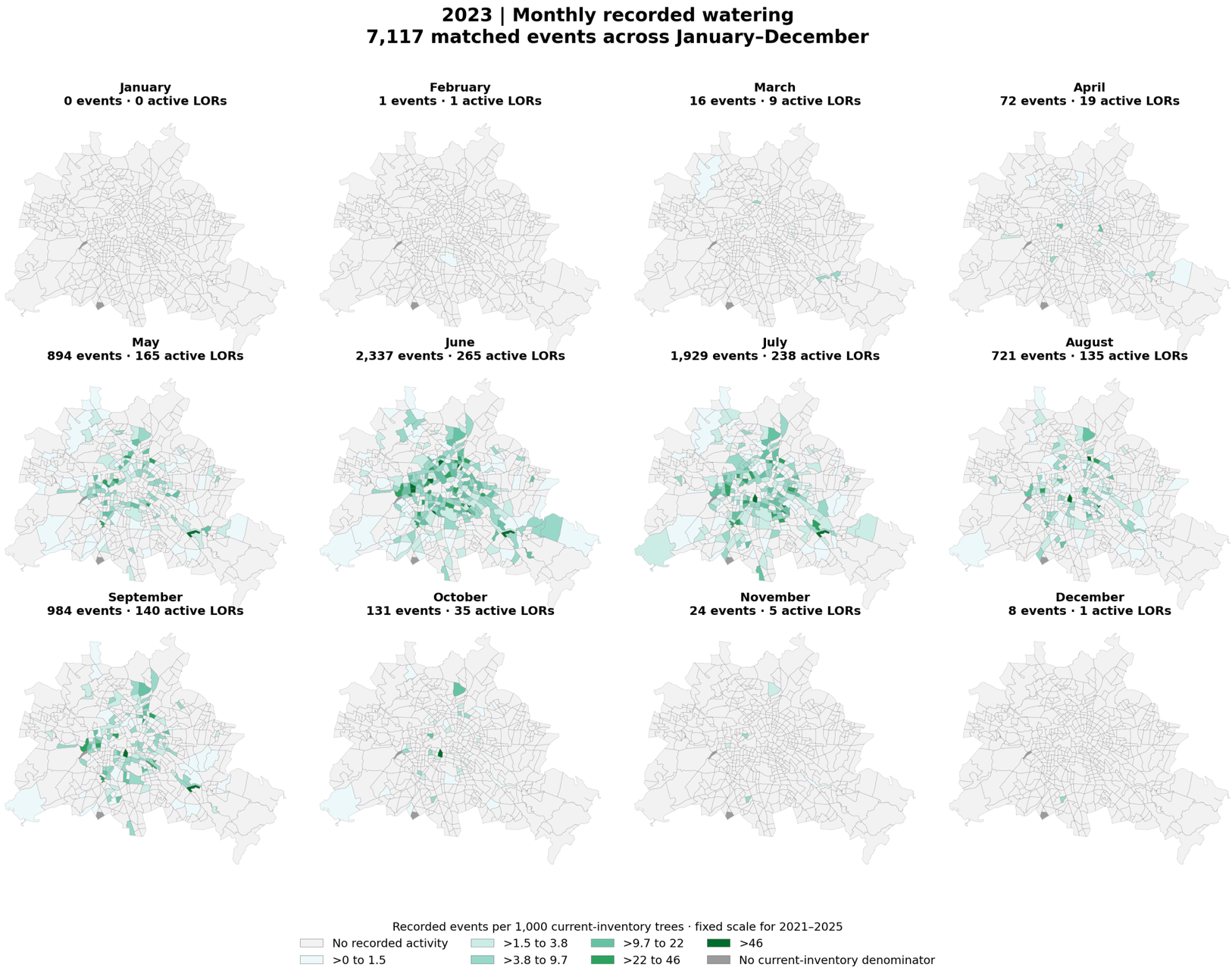
Twelve actual months; shared fixed classes across all years. Recorded platform activity only; current-inventory denominators; no inference about unlogged or municipal watering.

Data: data/processed/foundation/lor_month_recorded_watering_2020_2025.csv

Builder: pipelines/04_build_watering_temporal_atlas.py | Original: output/03_spatial_equity_vulnerability/maps/yearly/2022_monthly_recorded_watering_atlas.png

10 2023 monthly recorded watering

January-December 2023 | Monthly events per 1,000 current inventory trees



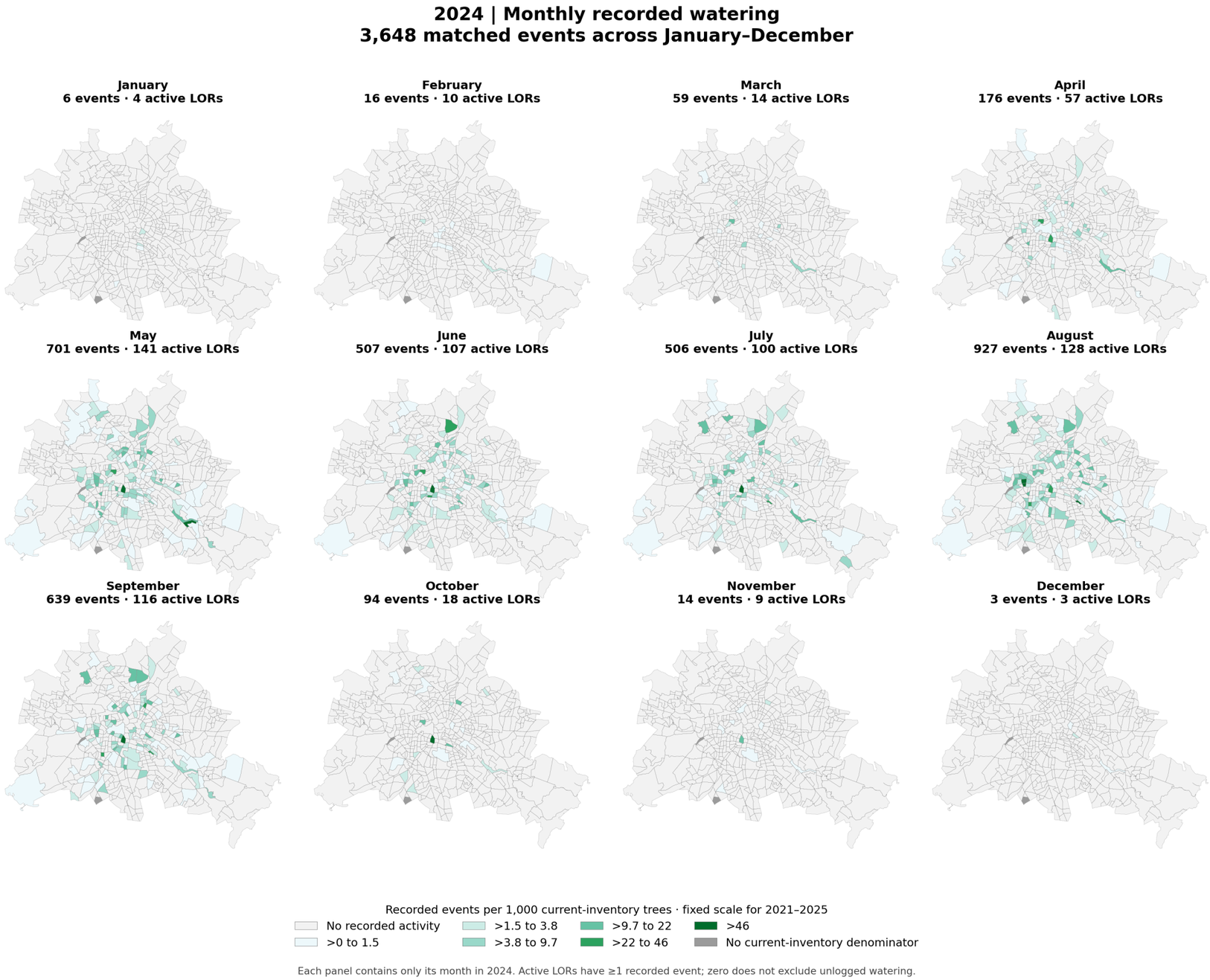
Twelve actual months; shared fixed classes across all years. Recorded platform activity only; current-inventory denominators; no inference about unlogged or municipal watering.

Data: data/processed/foundation/lor_month_recorded_watering_2020_2025.csv

Builder: pipelines/04_build_watering_temporal_atlas.py | Original: output/03_spatial_equity_vulnerability/maps/yearly/2023_monthly_recorded_watering_atlas.png

11 2024 monthly recorded watering

January-December 2024 | Monthly events per 1,000 current inventory trees



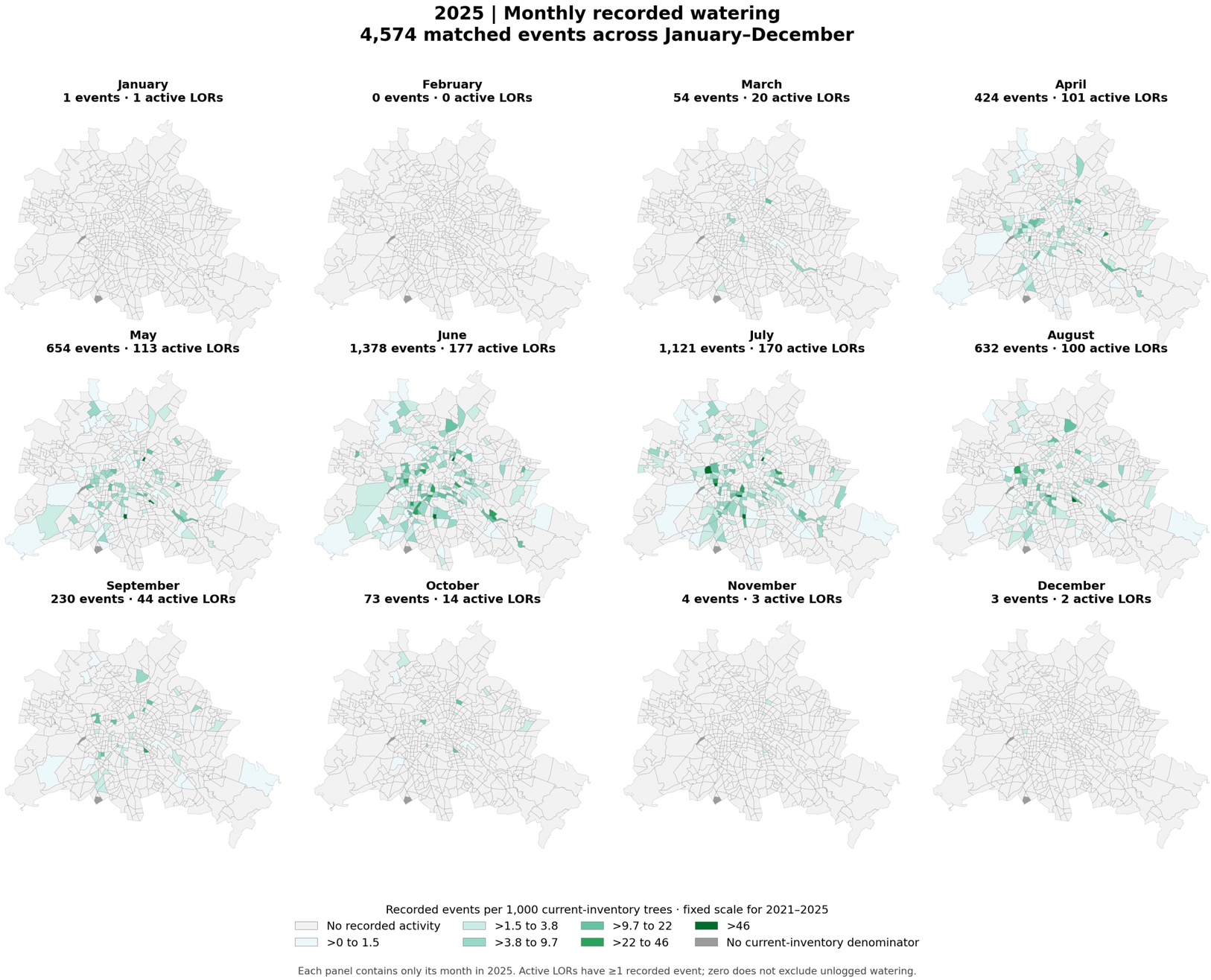
Twelve actual months; shared fixed classes across all years. Recorded platform activity only; current-inventory denominators; no inference about unlogged or municipal watering.

Data: data/processed/foundation/lor_month_recorded_watering_2020_2025.csv

Builder: pipelines/04_build_watering_temporal_atlas.py | Original: output/03_spatial_equity_vulnerability/maps/yearly/2024_monthly_recorded_watering_atlas.png

12 2025 monthly recorded watering

January-December 2025 | Monthly events per 1,000 current inventory trees



Twelve actual months; shared fixed classes across all years. Recorded platform activity only; current-inventory denominators; no inference about unlogged or municipal watering.

Data: data/processed/foundation/lor_month_recorded_watering_2020_2025.csv

Builder: pipelines/04_build_watering_temporal_atlas.py | Original: output/03_spatial_equity_vulnerability/maps/yearly/2025_monthly_recorded_watering_atlas.png